

Hongye Zu (she/her)

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EDUCATION

MSc. in Interactive Media, University College Cork, Ireland

Ireland

Thesis title: *Urban Light Pollution Visualization System*

Sept. 2024 – Oct. 2025

MSc. in Communication, Taylor's University

Malaysia

Thesis title: Vloggers' Self-Presentation on TikTok: Motives and Impact

Sept. 2020 – Aug. 2022

BSc. in Computer Science, Fujian Normal University

China

Sept. 2015 – Oct. 2019

RESEARCH PROJECT

Urban Light Pollution Visualization via Crowdsourced Smartphone Imaging

- Designed and built a native iOS crowdsensing application (Swift, AVCaptureSession) that captures night-time photos with automatic GPS tagging, timestamping, and on-device brightness estimation via perceptual grayscale weighting, paired with a Node.js/MongoDB backend and a Mapbox GL JS web interface for heatmap visualization.
- Piloted the system in Harbin, China, to test three research questions on the feasibility, spatiotemporal analytic value, and participatory challenges of smartphone-based light pollution monitoring.
- Conducted an 8-participant usability study, achieving a mean ease-of-use rating of 4.6/5 and average task completion time of 28 seconds; 7 of 8 participants found the heatmap visualization intuitive.
- Validated end-to-end system performance across 20 full-cycle tests (capture → upload → visualization), achieving an average latency of 8.7 seconds and sustained API throughput of ~110 requests/second.

Vloggers' Self-Presentation on TikTok: Motives and Impact

- Conducted a qualitative study applying Goffman's Dramaturgical Theory to examine how young Chinese TikTok vloggers construct self-presentation, using purposive sampling to recruit 4 vloggers (each active 1+ year with 3,000+ followers).
- Designed and conducted in-depth interviews and applied thematic analysis to code transcripts against three research questions covering self-presentation methods, motivation, and audience impact.
- Identified three core themes — stage-shaping behaviors, intrinsic/extrinsic motivations (social connection, monetization, self-realization), and dual positive/negative impacts of sustained self-presentation — and situated findings within the broader self-presentation and social media literature.
- Established analytical rigor through iterative cross-checking of interview data against theoretical literature and comparative analysis across respondents.

WORK EXPERIENCE

Store Manager & Social Media Coordinator (Part-time), Onekick Ireland

Mar. 2023 – Dec. 2024

- Managed daily store operations and social media content (TikTok, Instagram), driving a 200% year-over-year increase in Christmas season sales through coordinated marketing and team planning.

New Media Operations Assistant, Fuzhou Yizhong Media, China

Sept. 2018 – May 2019

- Produced and analyzed short-video and article content for WeChat official account operations, driving 25% follower growth and 40% higher average readership through data-informed publishing decisions.

AWARDS & SCHOLARSHIPS

- SEFS Merit Scholarship (€2,550), University College Cork, 2024
- Gold Award, “Challenge Cup” Student Competition, 2018
- Silver Award, “Internet+” Youth Innovation Competition, 2017

SKILLS & INTERESTS

Research Methods: User-Centred Design, Usability Testing, In-Depth Interviewing, Thematic Analysis, Purposive Sampling, Prototype Evaluation, Functional Testing, User Feedback Analysis, qualitative analysis

Technical: Swift, Xcode, Node.js, MongoDB, Mapbox GL JS, Python, JavaScript, HTML/CSS, C/C++, Git, R

Languages: English (Proficient), Chinese (Native)

Interests: Photography, Coffee Brewing, Traveling, Visiting Museums & Design Exhibitions, Everyday Interaction Design

PUBLICATION

Zu H., O'Mullane J., Urban Light Pollution Visualization via Crowdsourced Smartphone Imaging (In preparation)